

Individual Report

Task 4: Mining Insights from Customer Feedback

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EMATM0067 - Introduction to AI and Text Analytics

Group Git repository: <https://github.com/vd0208/Task4.git>

Personal branch: <https://github.com/vd0208/Task4/tree/Aanchala>

1 Individual Contribution Summary

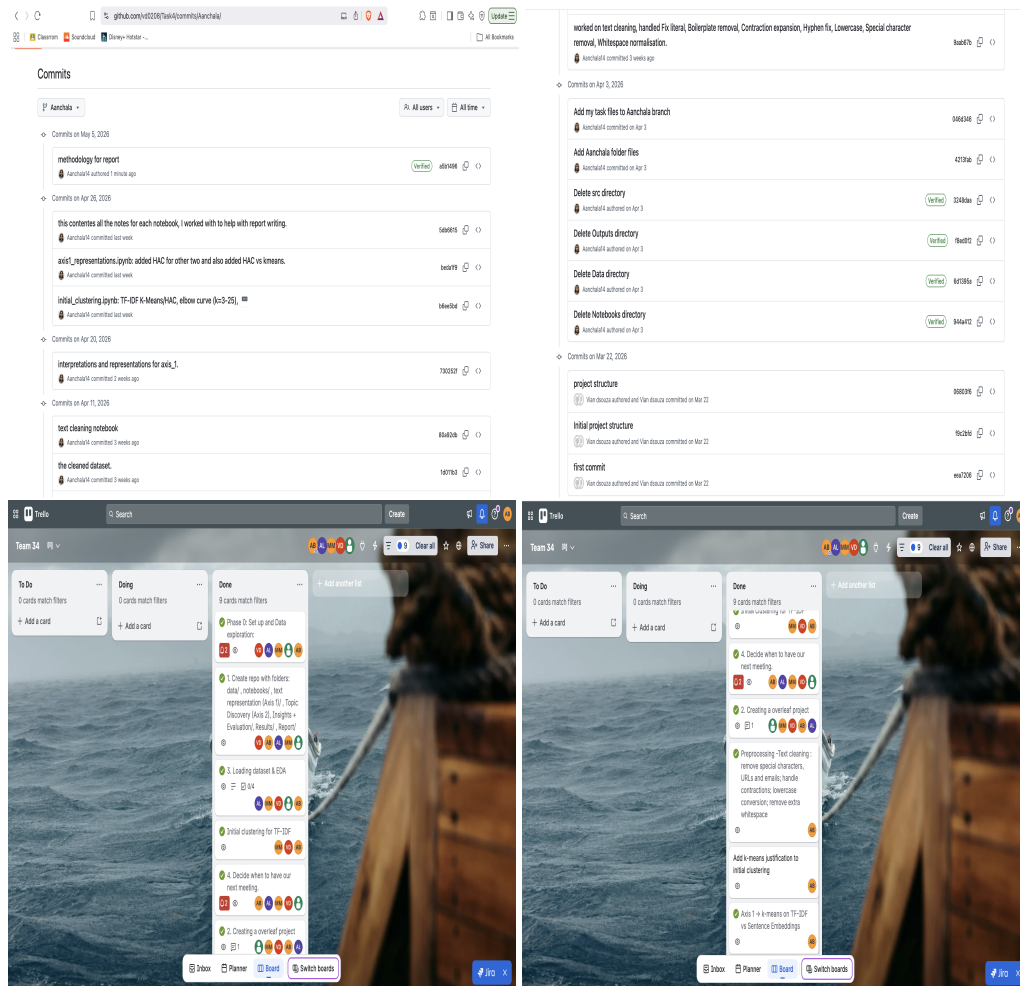


Figure 1: Top: GitHub commit history on the Aanchala branch showing commits from March to May 2026. Bottom: Trello board (Team 34) showing completed tasks assigned to me (AB), including EDA, text cleaning, initial clustering, and Axis 1.

My contributions spanned the full pipeline from raw data exploration to Axis 1 evaluation. I performed **Exploratory Data Analysis** alongside the team, with my EDA selected as the version carried forward to main. I implemented the complete **text cleaning pipeline** (literal newline removal, boilerplate stripping, contraction expansion, hyphen correction, lowercasing, special character removal). I did **initial clustering** on TF-IDF, including elbow curve analysis ($k \in \{5, 6, 10, 15\}$, five seeds), HAC dendrogram, K-Means vs HAC comparison, multi-seed stability analysis, and selection of $k = 5$. I implemented **Axis 1** comparing TF-IDF and sentence embeddings using both K-Means and HAC across all metrics. I also contributed to the **report abstract and methodology**.

2 Technical Understanding

2.1 Technical Challenge

The first difficulty was representation selection. TF-IDF was the obvious baseline, but choosing what to compare it against required careful thought. Sentence embeddings (all-MiniLM-L6-v2) were the natural modern alternative, but I also experimented with domain-trained Word2Vec, training vectors directly on the customer support corpus so the model could learn specialist vocabulary (e.g. medical-hospital-breach, analytics-investment-optimize).

Secondly, during Axis 1 evaluation, I had to determine how to meaningfully label and interpret clusters beyond simplistic positive/negative sentiment categories. Customer support tickets contain nuanced, domain-specific language, and initial attempts to describe clusters by sentiment alone failed to capture the actual nature of the issues customers were raising. My approach was to combine two complementary signals for each cluster. First, I extracted the top-10 highest-weighted TF-IDF terms from the cluster centroid, revealing the dominant vocabulary. Second, I identified the most representative ticket from each cluster by computing cosine distances between all ticket vectors and the centroid, selecting the ticket geometrically closest to it. This ensured the sample ticket was not a random example but the most typical one in the cluster. Together, these two signals allowed meaningful theme labels to be assigned: for example, a cluster dominated by terms such as *security*, *medical*, *breach*, *hospital* with a representative ticket describing unauthorised access to healthcare records was labelled “Healthcare & Security Issues”. This approach was applied consistently across representations in Axis 1, enabling qualitative comparison of how TF-IDF and sentence embeddings grouped tickets thematically.

A computational difficulty was while encoding 27,919 tickets with all-MiniLM-L6-v2 took over 10 minutes and repeatedly failed mid-run in the notebook environment. To resolve this, I serialised the computed embedding matrix to disk as `embeddings_minilm.npy` immediately after encoding, so all subsequent experiments, K-Means, HAC, t-SNE, silhouette plots, coherence scoring could reload the embeddings instantly rather than recomputing them.

2.2 Impact on Results

The cluster interpretation method had a direct impact on the group’s findings. It revealed that all representations consistently identified the same five customer issue themes - Technical Support & System Failures (44.0%), Healthcare & Security Issues (19.7%), SaaS & Project Management (13.4%), Digital Marketing & Brand Strategy (13.5%), and Data Analytics & Investment Decisions (9.3%) regardless of the underlying representation method. This cross-method consistency was itself a key finding: it demonstrated that these five issue categories reflect genuine structure in the data rather than artefacts of any single approach. The named cluster themes were also adopted directly in the group report and informed the actionable insights section, providing the business-facing framing of what types of problems customers most frequently raise. The $k = 5$ selection from initial clustering was adopted as the fixed cluster count throughout, ensuring methodological consistency across all experiments. A poorly justified k would have undermined every downstream comparison.

Axis 1 evaluation revealed an unexpected result: despite our hypothesis that sentence embeddings would outperform TF-IDF, TF-IDF achieved higher coherence (0.7605) and near-perfect stability while embeddings achieved a substantially higher silhouette score (0.1214 vs 0.0381). It provided the group with a richer and more honest evaluation, directly informing the trade-off discussion in the conclusions.

3 Teamwork

We (Team 34) coordinated work using GitHub for version control, with each member working on a dedicated branch and merging into main via pull requests. Trello was used for task assignment and progress tracking. WhatsApp was used for day-to-day communication and sharing results between sessions.

Roles were distributed as follows. Aanchala (myself) handled EDA (selected version), text cleaning, initial clustering, Axis 1 evaluation, and the report abstract and methodology section. Xiao led tokenisation and lemmatisation in preprocessing and contributed to Axis 1 evaluation metrics and the report conclusion. Annie handled EDA and preprocessing, led the report introduction, methodology and evaluation sections, and coordinated the overall report structure. Mengxiao Ma implemented TF-IDF vectorisation and BERTopic for Axis 2. Vian implemented sentence embeddings, LDA topic modelling, Axis 2 evaluation, and the methodology.